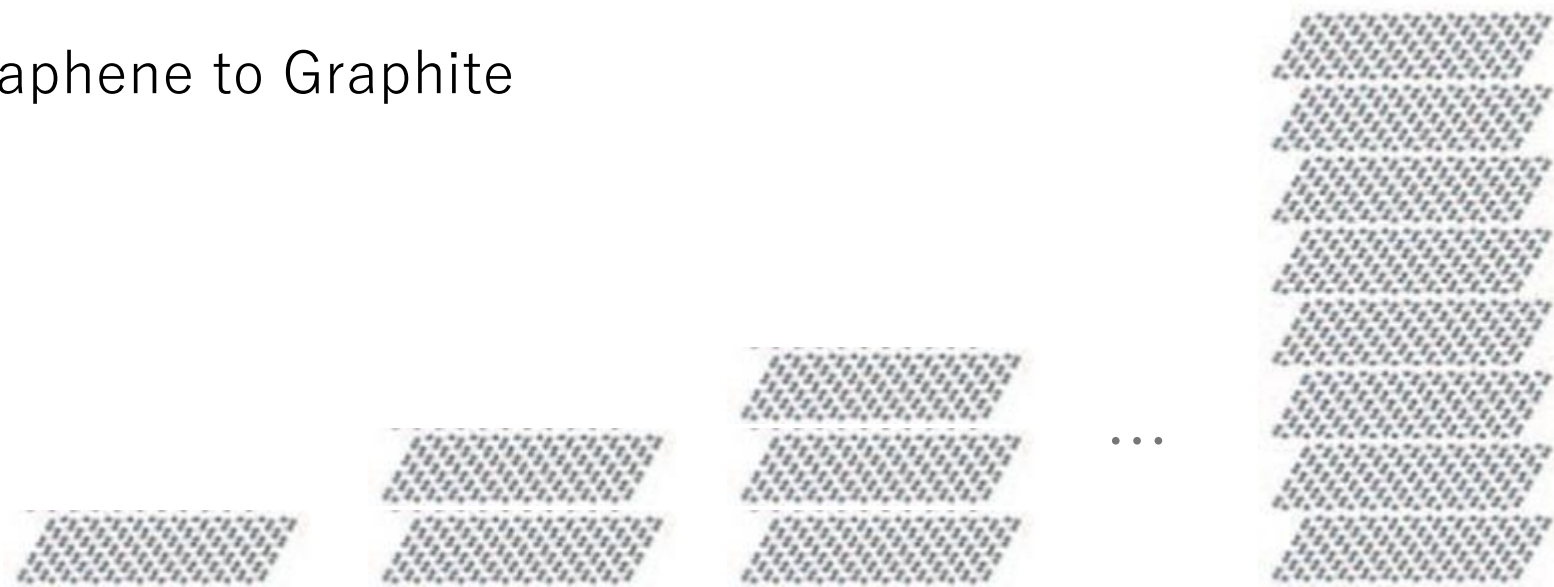
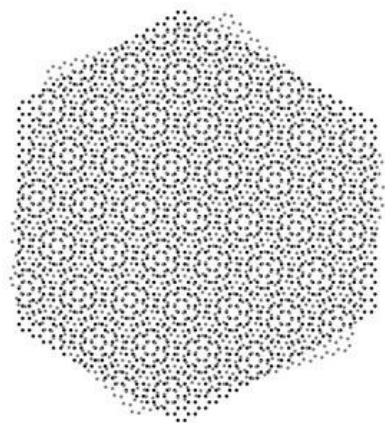


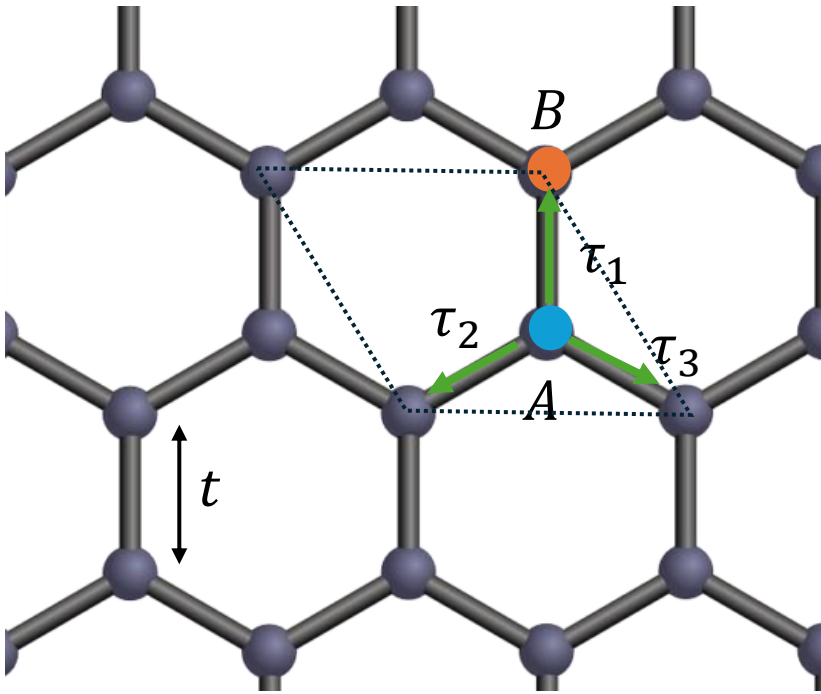
Graphene to Graphite



Twisted bilayer graphene



Graphene



Tight binding

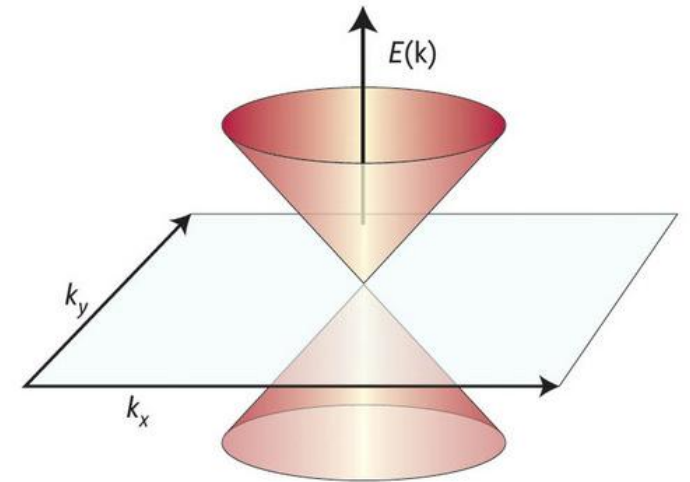
$$H = \begin{pmatrix} A & B \\ 0 & h \end{pmatrix} \begin{matrix} A \\ B \end{matrix}, \quad h = -t \sum_{j=1}^3 e^{-ik \cdot \tau_j}$$

When $k \simeq K$,

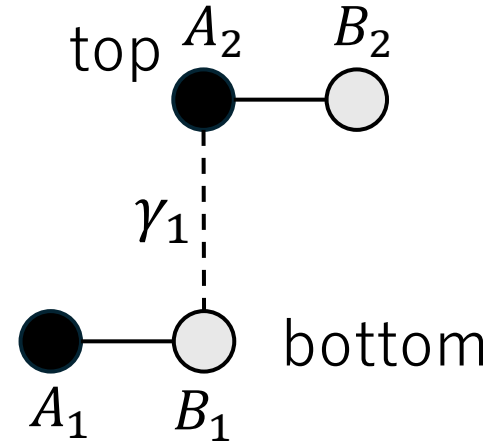
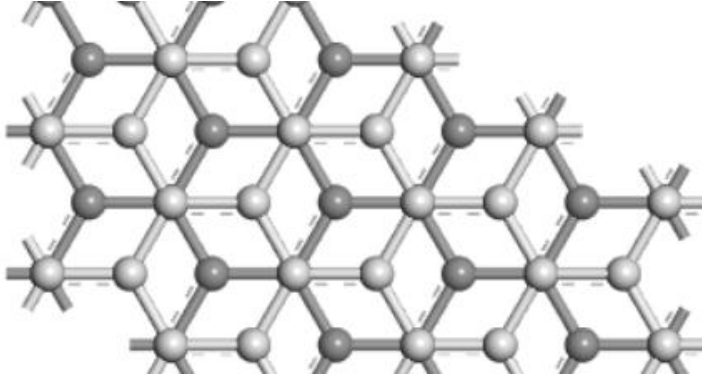
$$H = \hbar v \begin{pmatrix} 0 & k_x - ik_y \\ k_x + ik_y & 0 \end{pmatrix}$$

$$(v = \frac{\sqrt{3}at}{2\hbar})$$

$$\Rightarrow E = \hbar v k$$



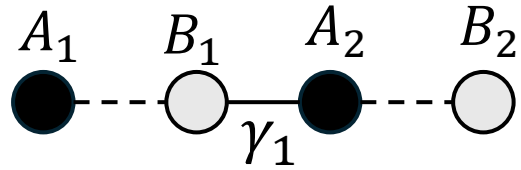
Bilayer graphene



Tight binding

$$H = \begin{pmatrix} \boxed{\begin{matrix} A_1 & H_{MLG} & B_1 \\ 0 & \hbar v(\pm k_x - ik_y) \\ \hbar v(\pm k_x + ik_y) & 0 \end{matrix}} & \boxed{\begin{matrix} A_2 & V & B_2 \\ 0 & 0 \\ \gamma_1 & 0 \end{matrix}} \\ \boxed{\begin{matrix} 0 & \gamma_1 \\ 0 & 0 \end{matrix}} & \boxed{\begin{matrix} 0 & \hbar v(\pm k_x - ik_y) \\ \hbar v(\pm k_x - ik_y) & 0 \end{matrix}} \end{pmatrix} \begin{matrix} A_1 \\ B_1 \\ A_2 \\ B_2 \end{matrix}$$

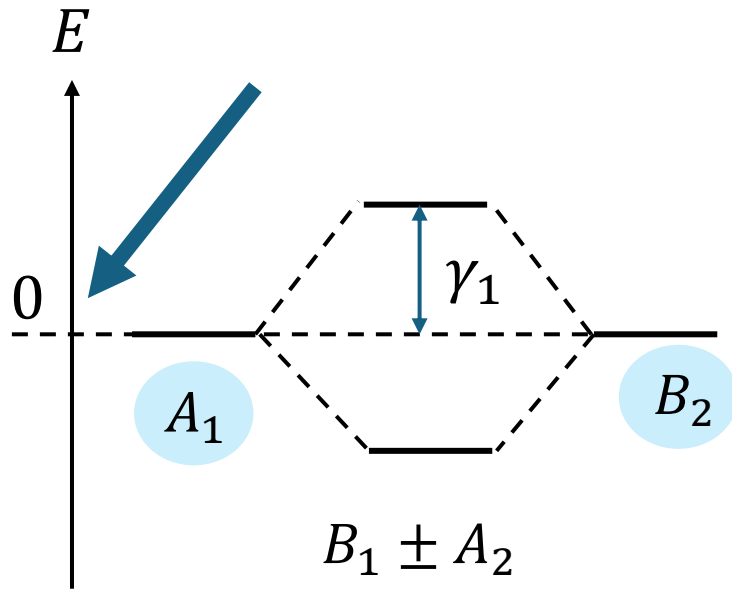
Bilayer graphene: low energy Hamiltonian



$$H = \begin{pmatrix} A_1 & B_2 & B_1 & A_2 \\ 0 & 0 & \hbar v(\pm k_x - i k_y) & 0 \\ 0 & 0 & 0 & \hbar v(\pm k_x + i k_y) \\ \hbar v(\pm k_x + i k_y) & 0 & 0 & \gamma_1 \\ 0 & \hbar v(\pm k_x - i k_y) & \gamma_1 & 0 \end{pmatrix} \begin{pmatrix} A_1 \\ B_2 \\ B_1 \\ A_2 \end{pmatrix}$$

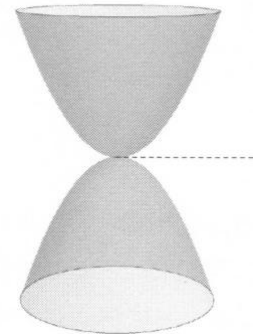
$$\equiv \begin{pmatrix} H_1 & V^\dagger \\ V & H_2 \end{pmatrix}$$

Derive the effective Hamiltonian in the (A_1, B_2) basis

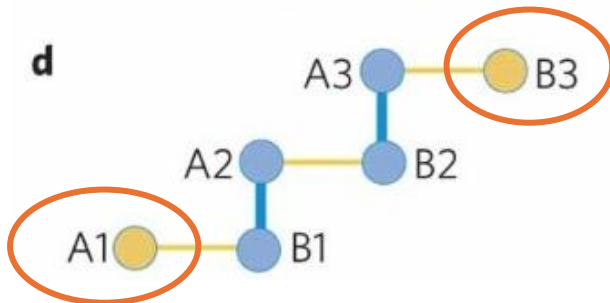
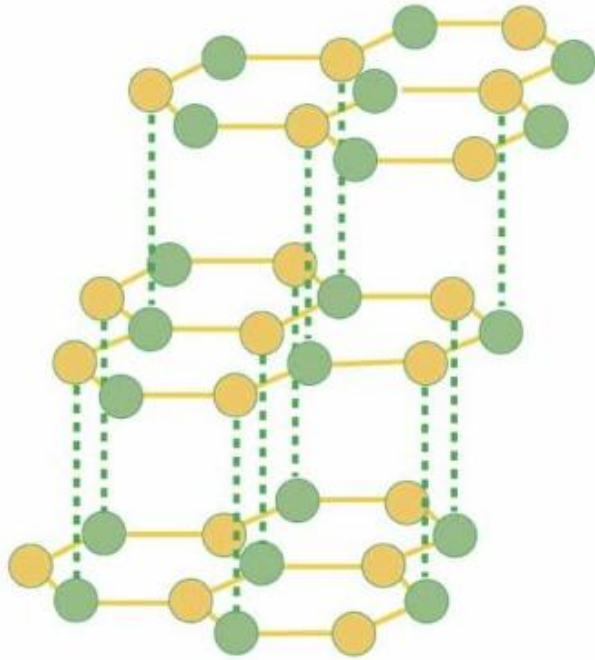


$$H_{\text{eff}} = H_1 - V^\dagger H_2 V = -\frac{\hbar^2 v^2}{\gamma_1} \begin{pmatrix} A_1 & B_2 \\ 0 & (\pm k_x - i k_y)^2 \\ (\pm k_x + i k_y)^2 & 0 \end{pmatrix} \begin{pmatrix} A_1 \\ B_2 \end{pmatrix}$$

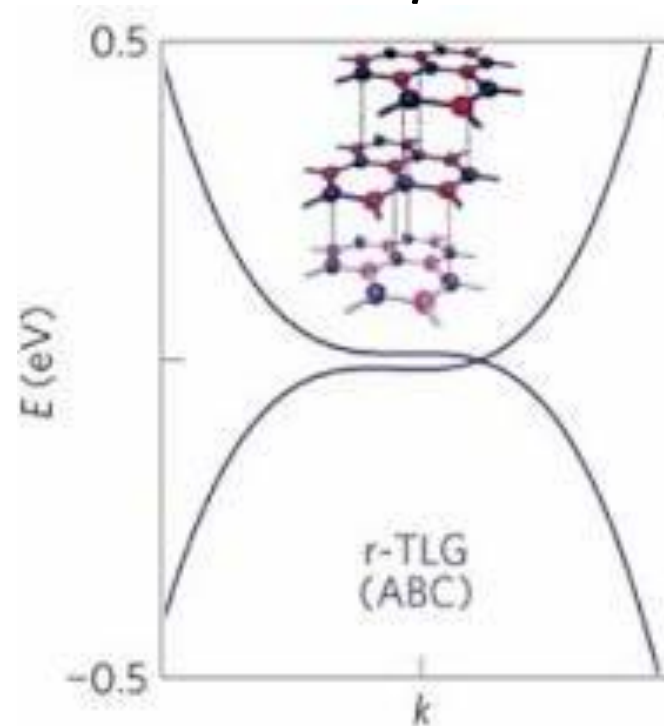
$$\Rightarrow E = \frac{(\hbar v k)^2}{\gamma_1}$$



ABC trilayer graphene metastable



$$E = \pm \frac{(\hbar v k)^3}{\gamma^2}$$



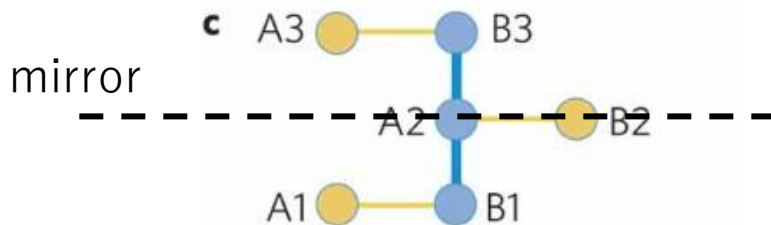
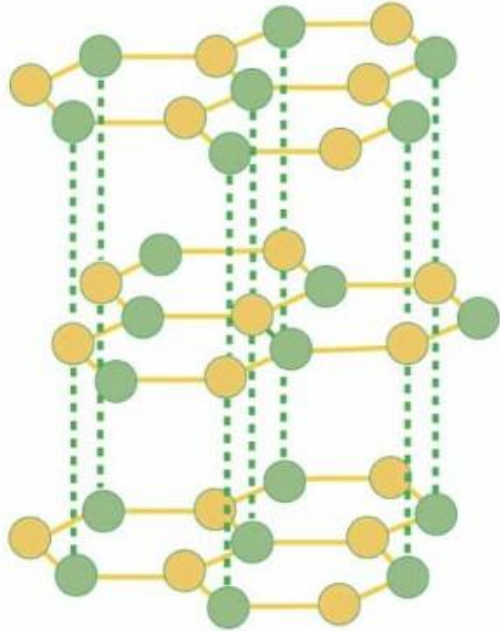
N-layer rhombohedral graphene

$$E = \pm \frac{(\hbar v k)^N}{\gamma^{N-1}}$$

Superconductivity,
Fractional quantum Hall,,,

ABA-Trilayer graphene stable

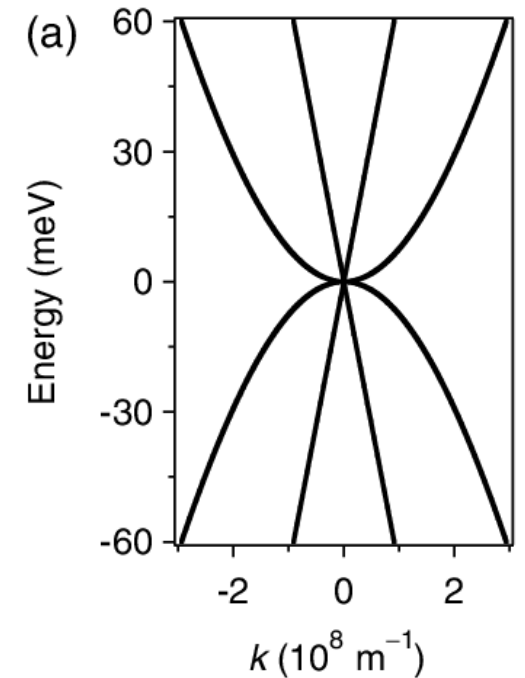
a ABA trilayer



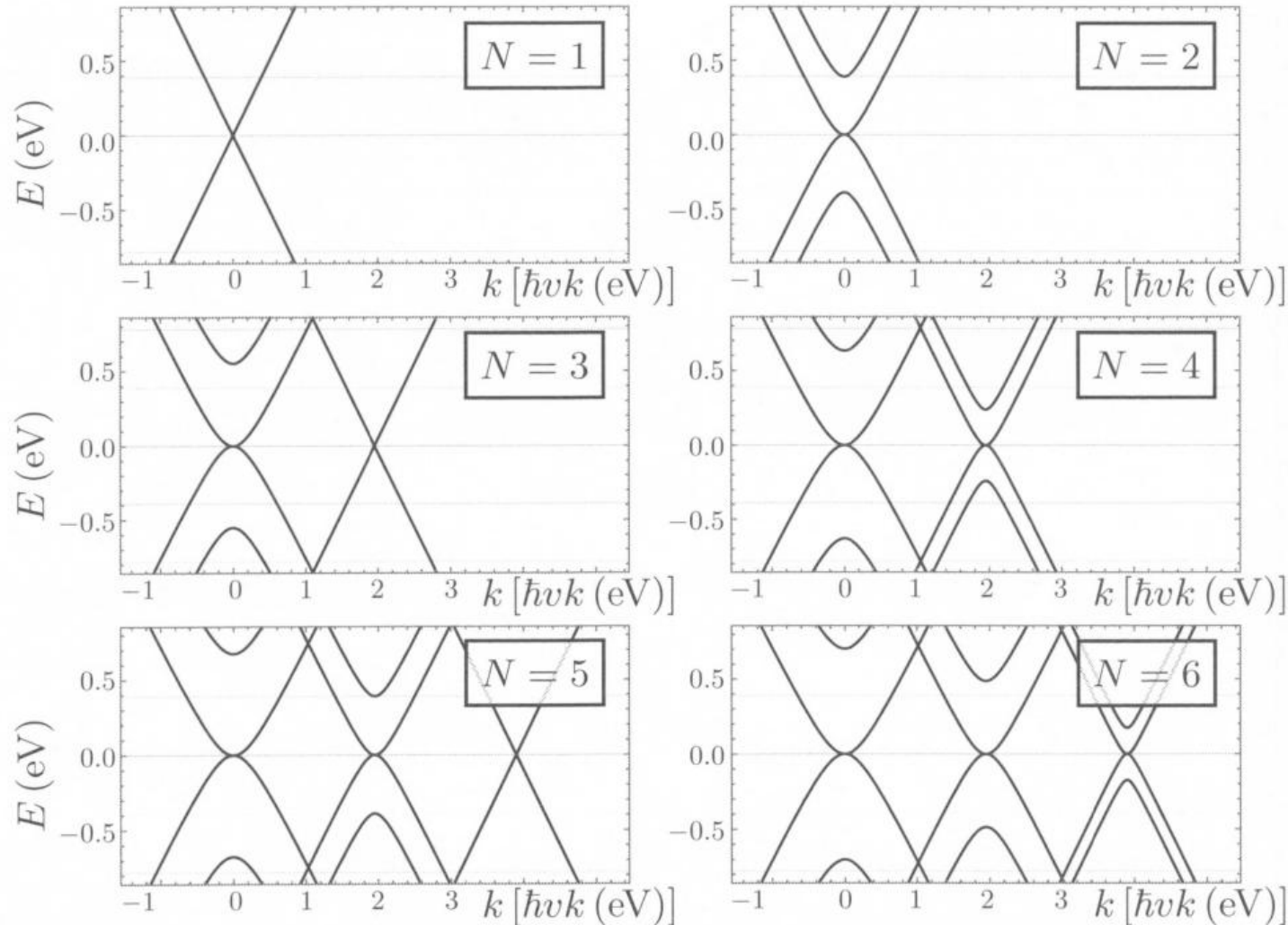
$$H = \begin{pmatrix} |1\rangle & |2\rangle & |3\rangle \\ H_{MLG} & V & 0 \\ V^\dagger & H_{MLG} & V^\dagger \\ 0 & V & H_{MLG} \end{pmatrix}$$

$$U^\dagger H U = \begin{pmatrix} |1\rangle + |3\rangle & |2\rangle & |1\rangle - |3\rangle \\ \boxed{H_{MLG}} & \sqrt{2}V & 0 \\ \sqrt{2}V^\dagger & \boxed{H_{MLG}} & 0 \\ 0 & 0 & \boxed{H_{MLG}} \end{pmatrix}$$

Bilayer Monolayer

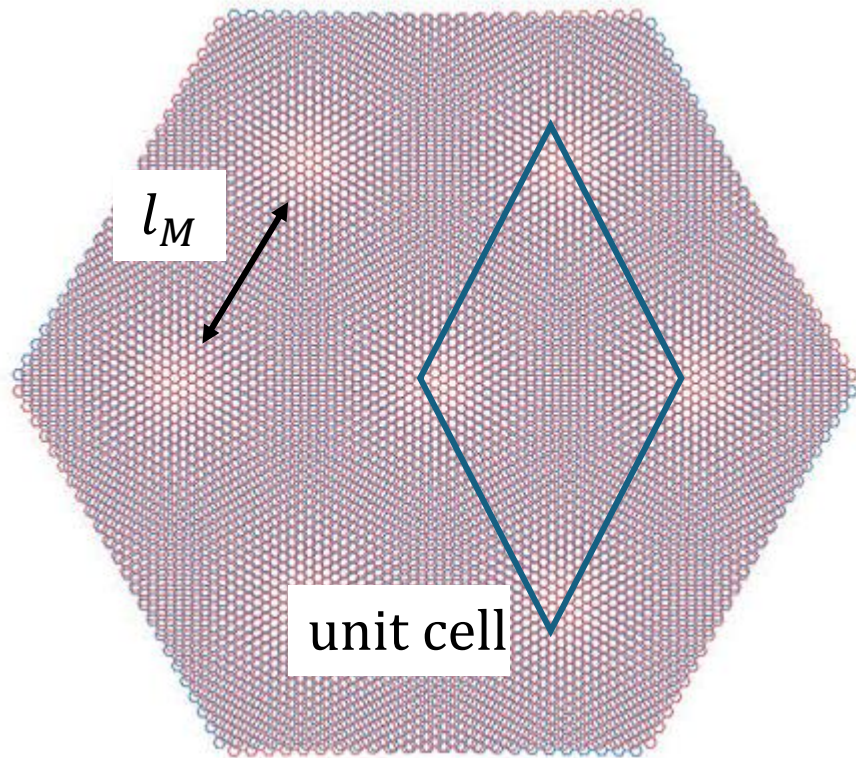


Bernal N-layer graphene



1. Monolayer
2. Bilayer
3. Monolayer + Bilayer
4. Bilayer + Bilayer
5. Monolayer + Bilayer + Bilayer
- ⋮

Twisted bilayer graphene



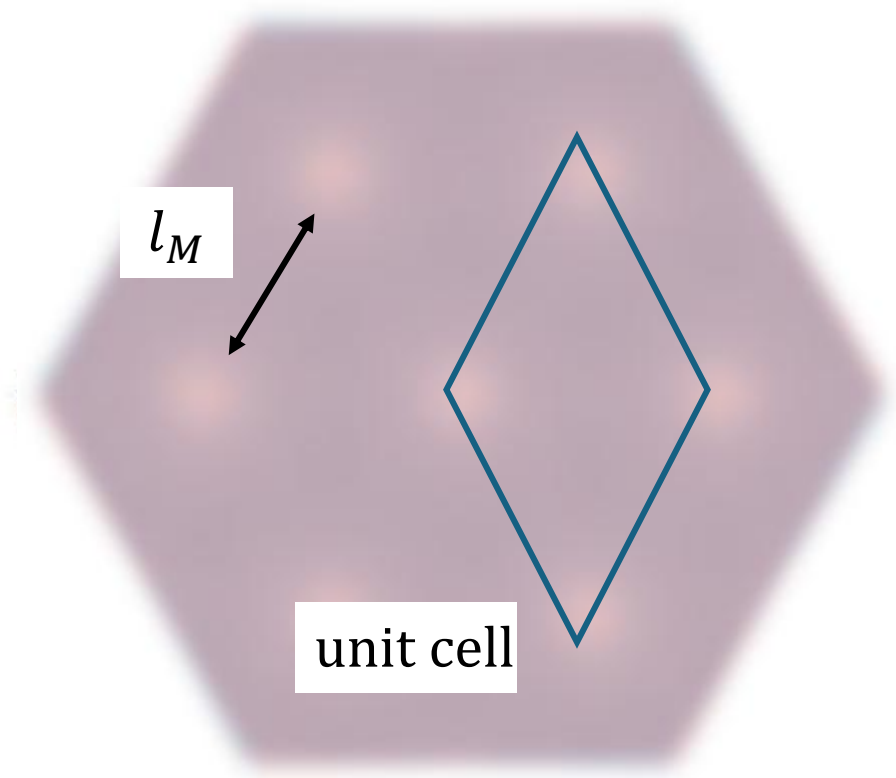
- Not periodic
 - atoms $> 10,000$ in a unit cell
- $$l_M = \frac{a}{2\sin\theta} \sim 14 \text{ nm } (\theta = 1.1^\circ)$$

Tight-binding model

$$H = \left(\begin{array}{c} \sim 10,000 \times 10,000 \end{array} \right)$$

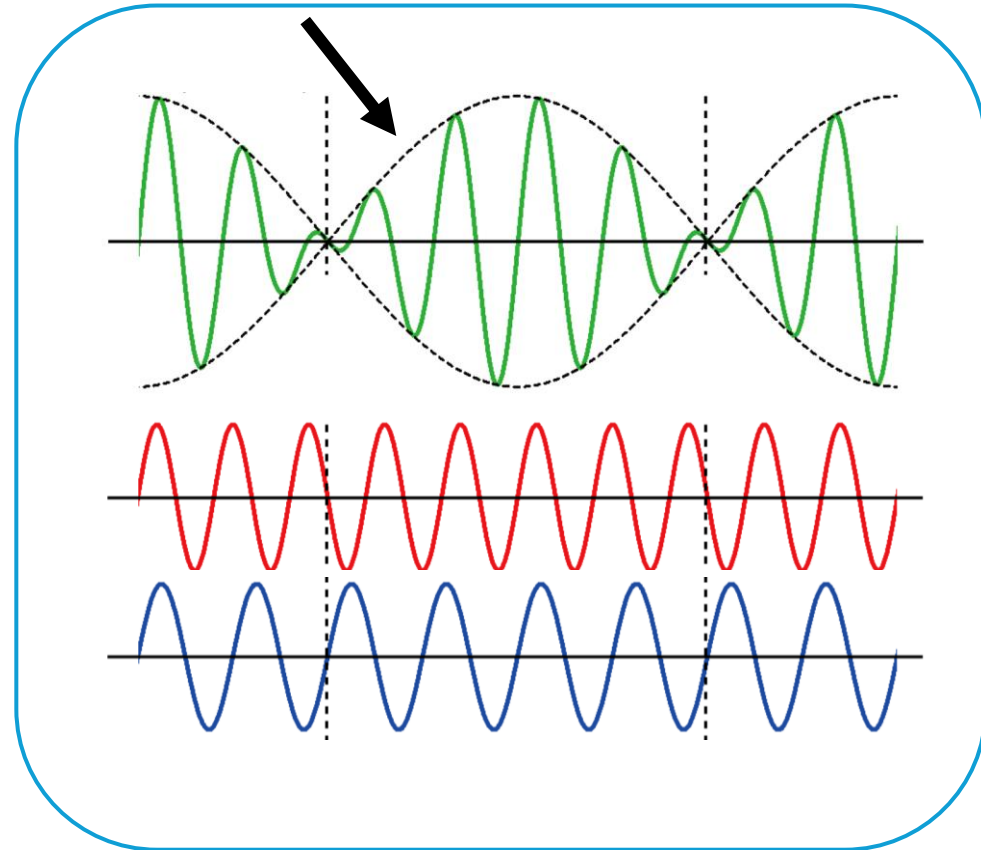
➤ Continuum model

Continuum model

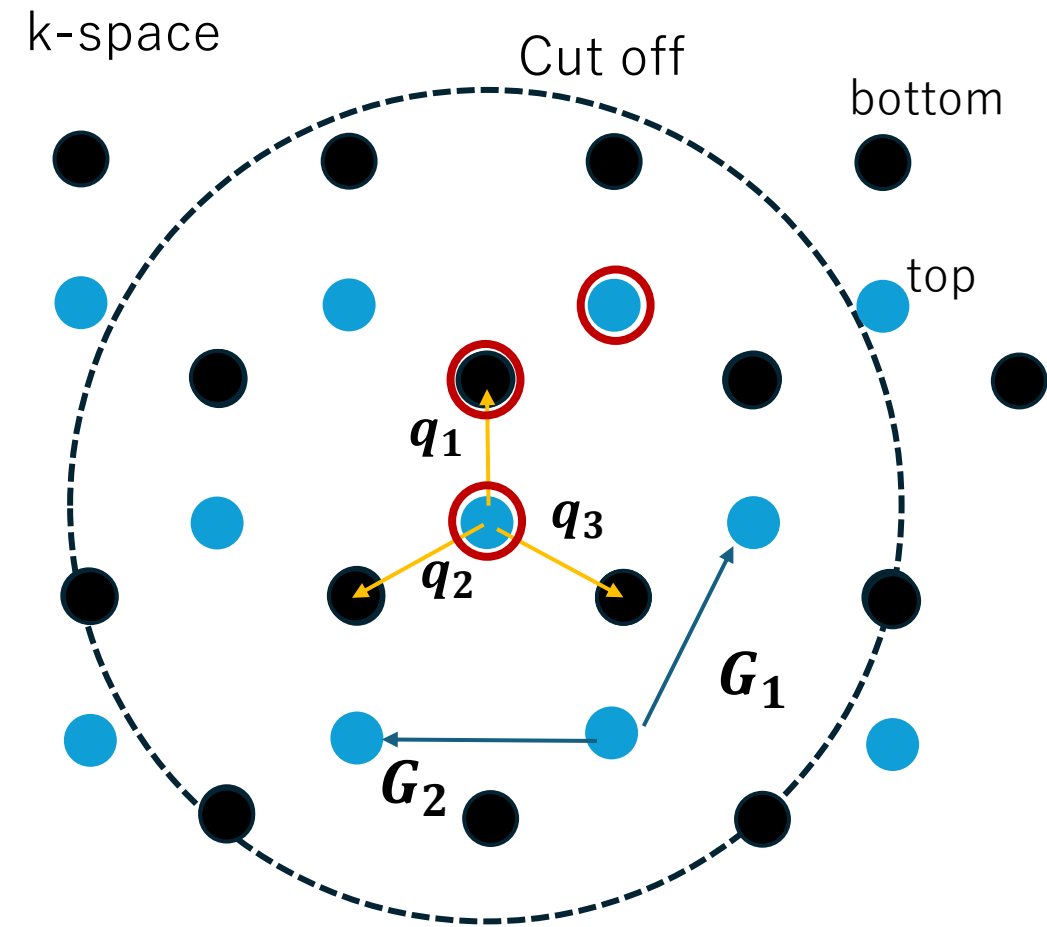


$$H = \begin{pmatrix} \overset{\text{top}}{\downarrow} H_{1,MLG} & U^\dagger \\ U & \overset{\text{bottom}}{\uparrow} H_{2,MLG} \end{pmatrix}, U = \sum_{j=1}^3 u T_j e^{i q_j \cdot r}, T_j = \begin{pmatrix} 1 & e^{-i \frac{2\pi}{3}(j-1)} \\ e^{i \frac{2\pi}{3}(j-1)} & 1 \end{pmatrix}$$

Interlayer hopping



Continuum model



Interlayer hopping

$$U = \sum_{j=1}^3 u T_j e^{i q_j \cdot r}, T_j = \begin{pmatrix} 1 & e^{-i \frac{2\pi}{3}(j-1)} \\ e^{i \frac{2\pi}{3}(j-1)} & 1 \end{pmatrix}$$

$$|k\rangle \xrightarrow{T_1} |k + q_1\rangle \xrightarrow{T_2} |k + \frac{q_1 - q_2}{G_1}\rangle \rightarrow \dots$$

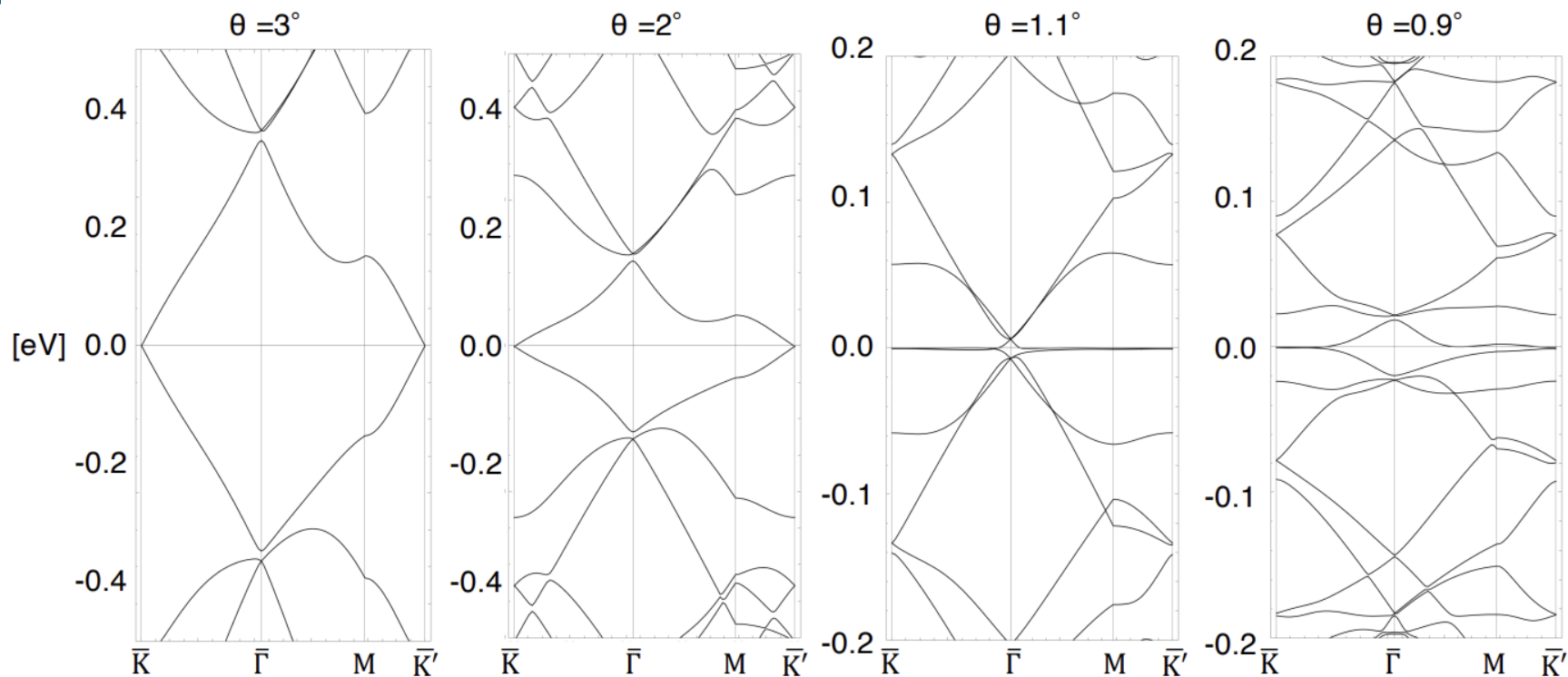
Cut off

$$H = \begin{pmatrix} H_{MLG}(k) & & & T_1^\dagger & T_2^\dagger \\ & H_{MLG}(k + G_1) & & & \\ & & \ddots & & \\ T_1 & T_2 & & H_{MLG}(k + q_1) & \\ & & & H_{MLG}(k + q_1 + G_1) & \ddots \end{pmatrix}$$

$$2(\text{cutoff}) \times 2(\text{cutoff}) \ll 10,000 \times 10,000$$

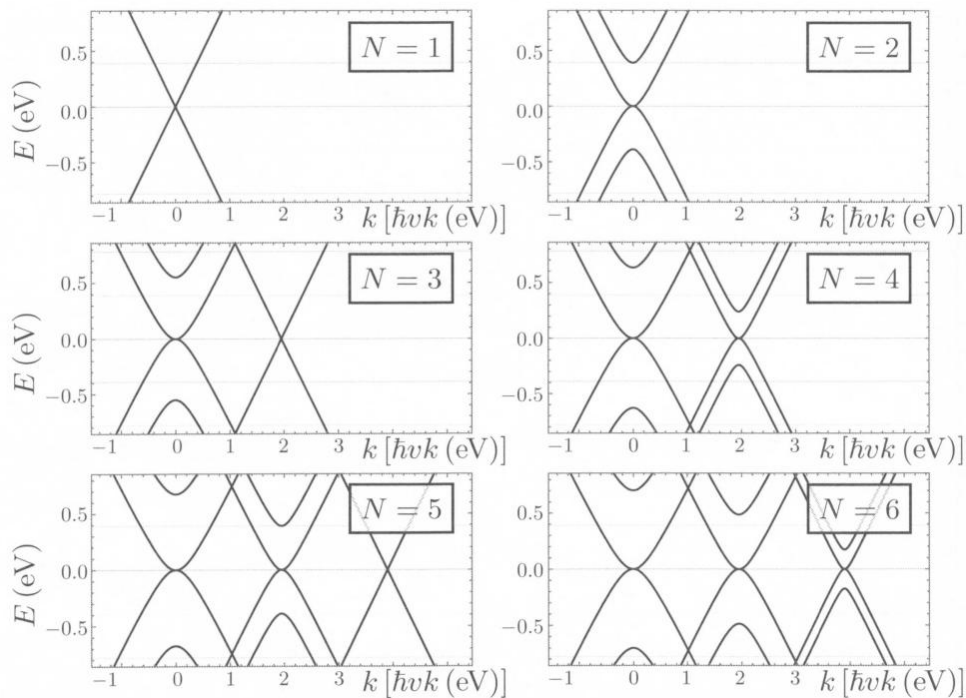
Twisted bilayer graphene

$$H = \left(\begin{array}{ccc|cc} H_{MLG}(k) & & & T_1^\dagger & \\ & H_{MLG}(k + G_1) & & T_2^\dagger & \\ & & \ddots & & \\ T_1 & T_2 & & H_{MLG}(k + q_1) & \\ & & & & H_{MLG}(k + q_1 + G_1) \end{array} \right)$$

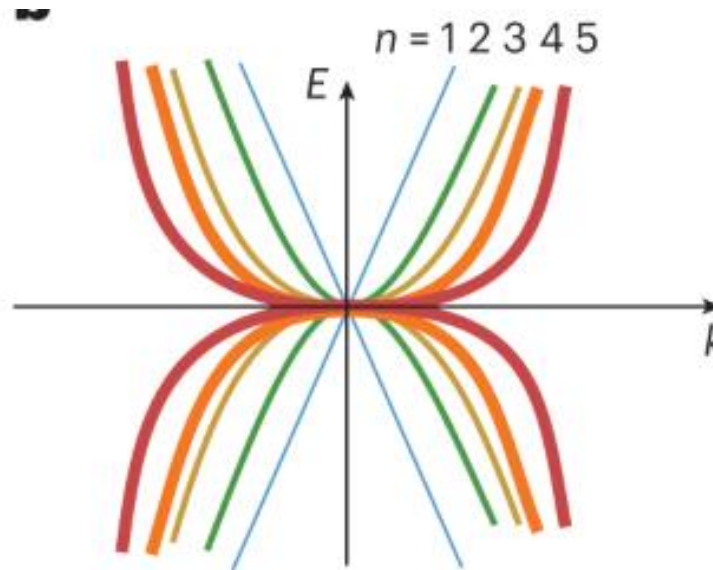


Summary

Bernal N-layer graphene

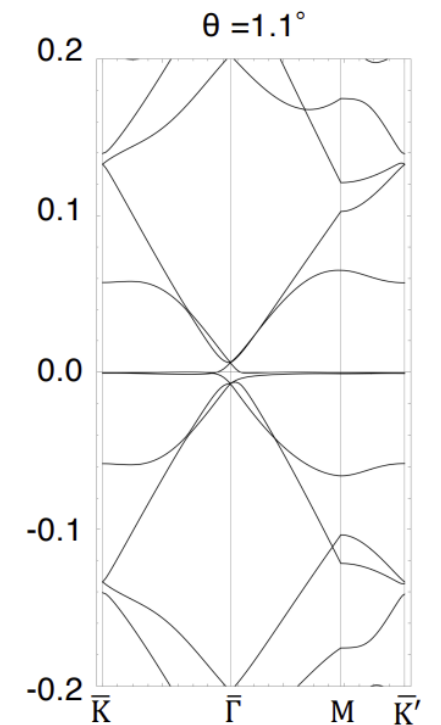


Rhombohedral graphene
 $E \sim k^N$

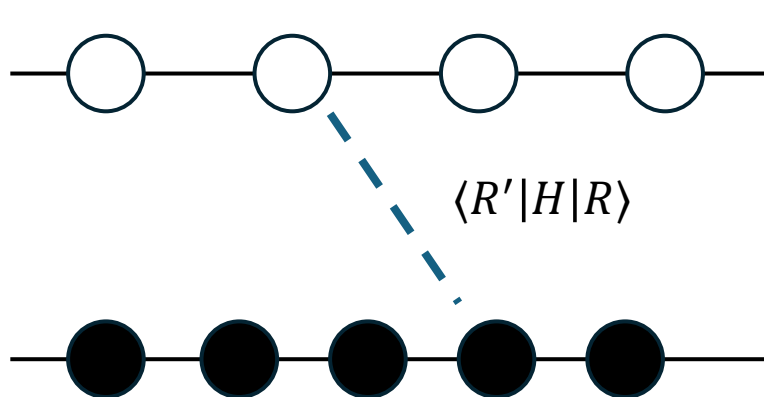


Twisted bilayer graphene

$$H = \begin{pmatrix} h_{top} & T^\dagger \\ T & h_{bottom} \end{pmatrix}$$



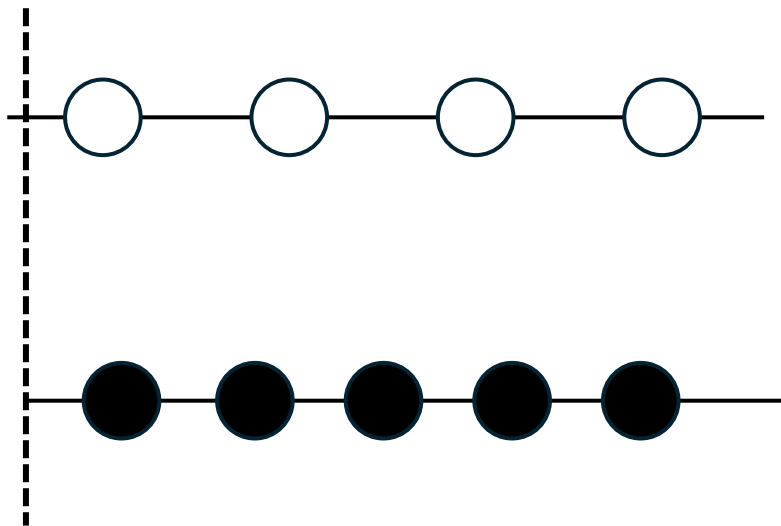
1D double chain



$$|k\rangle = \frac{1}{\sqrt{N}} \sum_R e^{ikR} |R\rangle$$

$$|k'\rangle = \frac{1}{\sqrt{N'}} \sum_{R'} e^{ik'R'} |R'\rangle$$

$$\begin{aligned} \langle k'|H|k\rangle &= \frac{1}{\sqrt{NN'}} \sum_{R,R'} \langle R'|H|R\rangle e^{-ik'R'} e^{ikR} \\ &= -\frac{1}{\sqrt{NN'}} \frac{\sqrt{aa'}}{L} t(q) e^{i(q-k')R'} e^{i(k-q)R} \\ &= -\sum_{G,G'} t(k+G) \delta_{k+G,k'+G'} \end{aligned}$$



$$\langle k' | H | k \rangle = \frac{1}{\sqrt{NN'}} \sum_{R,R'} \langle R' | H | R \rangle e^{-ik'(R'+\tau')} e^{ik(R+\tau)}$$

$$= - \sum_{G,G'} t(k+G) \delta_{k+G,k'+G'} e^{-iG\tau+iG'\tau'}$$

2D

$$\langle \mathbf{k}', X' | H | \mathbf{k}, X \rangle = - \sum_{\mathbf{G}, \mathbf{G}'} t(\mathbf{k} + \mathbf{G}) \delta_{\mathbf{k} + \mathbf{G}, \mathbf{k}' + \mathbf{G}'} e^{-i\mathbf{G} \cdot \boldsymbol{\tau}_X + i\mathbf{G}' \cdot \boldsymbol{\tau}_{X'}}$$

